UNIT-2

→ The 5G wireless propagation channels:

- Channel model requirements
- Propagation scenarios & challenges in the 5G modeling
- Channel Models for mmWave
- MIMO Systems

*5G Wireless Propagation Channel

→ 5G wireless Propagation channel is the Path / Physical medium. through which a 5G signal travels from transmitter to receiver.

⇒ 5G signal travelling transmitter (base station) to receiver (user equipment).

→ It describes how 5G signals move in air & how the environment affects them.

During Propagation

→ When a 5G signal travels, it may undergo:

- Reflection
- Diffraction
- Scattering
- loss of Power
- Delay Spread.

↳ these effects decide signal quality & speed.

Characteristics of 5G Propagation channels

→ High operating frequency

→ Short wavelength

→ High Path loss

→ Strong Sensitivity

→ Directional transmission

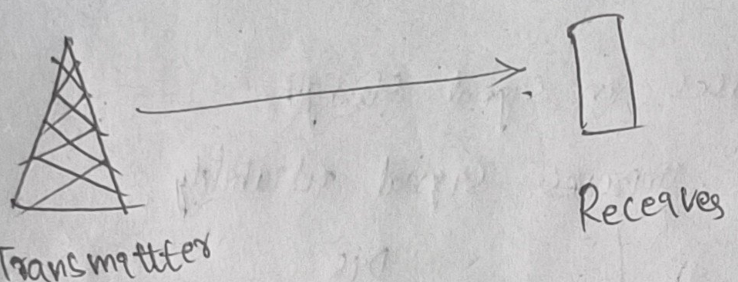
→ Rapid signal fading

* 5G Wireless Propagation Channel

→ When you use mobile:

- Tower sends signal
- Signal travels through air
- Reaches your phone.

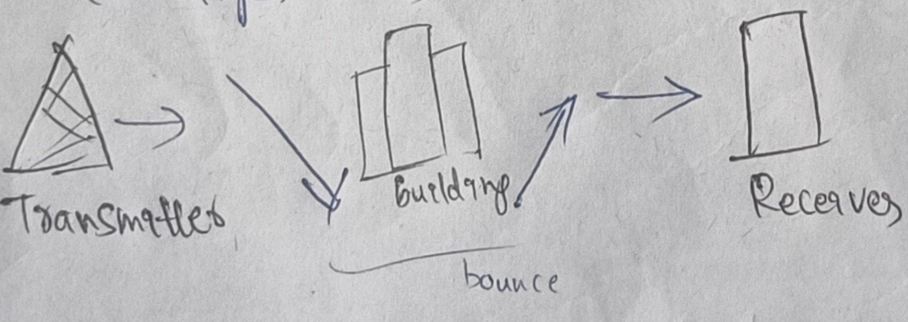
⇒ That whole path is Propagation channel.



Propagation Mechanism

1. Reflection

→ Signal bounces from large surfaces (buildings, walls)

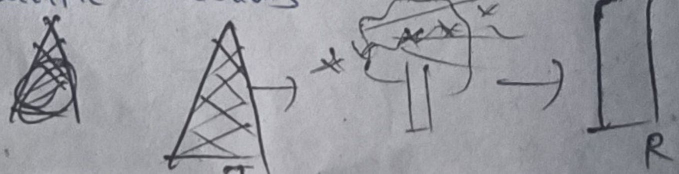


2. Diffraction

→ Signal bends

3. Scattering

→ Signal spreads in multiple directions (trees, Poles)



Types of 5G wireless Propagation channels

1. Line-of-Sight (LOS) channel :

→ A Propagation channel where a direct, unobstructed Path exists b/w transmitter & receiver.

- No blockage (No obstacles)
- minimum signal loss
- Strong received signal (strong signal)
- High data rate (High speed)

— Best performance

ex: ① 5G small cell to mbl in open ground, ③. open grou
② outdoor mmWave communication
④. Tower to mobile with clear view

2. Non-Line-of-Sight (NLOS) Channel :

→ A ~~Propo~~ Propagation channel where direct path is blocked by obstacles b/w transmitter & receiver.
- & signal reaches receiver through reflections.

- Buildings & walls block signals
- Higher path loss
- Reduced signal quality
- Signal reflects from objects.

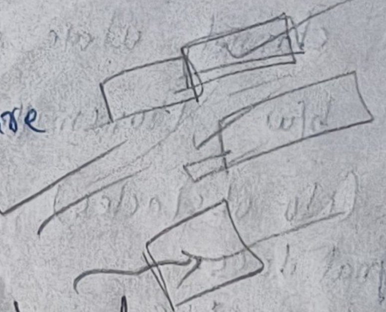
- Less speed
- weaker signal

⇒ Signal travels without direct Path
- due to obstacles.

3. Indoor Propagation channel

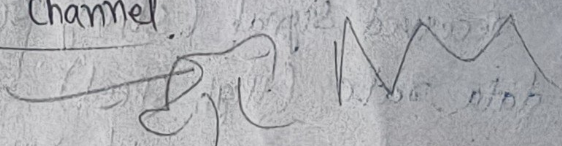
→ A channel where signal propagates inside buildings.

- Wall penetration loss
- Reflection from furniture



ex: 5G Wi-fi in offices

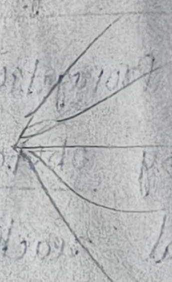
4. Outdoor Propagation channel



5. Multipath Propagation

→ Signal reaches receives through multiple paths.

- Same signal arrives many times
- causes delay
- signal reflects from walls
- common in cities.



ex: mbl used near buildings

6. Free space Propagation

→ Signal travels in open space without

- No reflection
- min loss

ex: Satellite communication.

* Propagation Channel Scenarios

→ Scenario means where & how the signal is used.

1. Indoor Scenario

→ Signal travels inside buildings

- walls, doors affect signal
- Short distance
- More reflections

ex:
- classroom
- Office
- Shopping mall

~~2. Outdoor~~

2. Urban Macrocell Scenario (UMa)

→ Base station placed above rooftop level in dense urban areas

- High building density
- Severe Shadowing

3. Urban Microcell Scenario (UMi)

→ Base station placed below roof top level, close to users.

- short cell range
- High Pb of LOS
- Strong reflections
- used for dense user areas.

ex: 5G Small cells in streets & markets.

⇒ PC seen Scenarios define where & how signals travels

4. Different environments.

→ 5G signals behave differently in different places.

ⓐ. Urban

- Area with many buildings
- ex: cities, towns
- Problems:
  - buildings
  - reflections

ⓑ. Suburban

- Area b/w city & village
- urban + rural

ⓒ. Rural

- open areas
- very few obstacles

Challenges

→ High path loss
→ Blockage sensitivity
→ Penetration loss
→ Doppler Effect
→ Environmental Effects.

② Indoor

- inside buildings
- walls, furniture block signals.

ex: 5G inside classrooms, malls.

* Channel Model → mathematical representation of wireless channel.

→ A channel model explains how a wireless signal travels from transmitter to receiver.

→ It is a mathematical and logical representation of how a wireless signal travels from T to R

RAFSS

Requirements

1. **Realism** :

→ The channel model should behave like a real wireless environment.

→ Model should behave like real signal.

→ Not imaginary or fake.

2. **Accurate** → should give precise results.

→ Model should give correct values.

→ Close to real measurements.

ex: If real signal is weak, model should not show strong signal.

3. **Flexibility** : → applicable to diff scenario (urban, rural)

→ It can be used in diff situations.

- works in indoor & outdoor
- works in city & village
- diff frequencies

4. Scalability :- → works for small to large n/ws

→ works for small & big n/ws
→ supports :- few users
- many users
- large n/ws

5. Standardization: → must follow global standards.

- everyone follows same rules
- defined by 3GPP

* mmWave - Millimete Wave

→ mmWave is a type of electromagnetic wave.

→ mmWave frequency range (30 GHZ - 300GHZ)

→ & wavelength b/w 1mm to 10mm

→ It is a high-frequency radio wave.

→ It is used in 5G

→ ultra-fast commⁿ

$\boxed{\text{* mmWave = High frequency + High Speed + Short range}}$

→ It has high data rate

→ Short wavelength

→ high frequency

→ uses beamforming

→ Limited coverage (range is very short)

→ ~~Ultra~~ Reduced interference.

→ Enables/supports new technologies like:

- VR/AR
- Autonomous Vehicle.

Advanta

• Disadvantages

- High Path loss (signal strength reduces)

- Poor Penetration
  Can't Pass through: wall, Tree
- Blockage Problem

- Short coverage.

Solutions to mmWave challenges

1. Beamforming

→ focuses signal in one direction

→ Improves signal strength.

2. Massive MIMO

→ uses multiple antennas.

3. Hybrid N/ws

Applications

- Smart cities
- IoT applications
- 5G mobile n/w
- Autonomous vehicles
- HD video streaming.

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* MIMO - Multiple Input Multiple Output

→ MIMO is a wireless commⁿ technique

→ It uses multiple antennas at both transmitter & receiver.

  - to improve data rate & reliability.

⇒ Instead of one antenna → we use many antennas

⇒ So more data will be sent at the same time.

Types of MIMO System

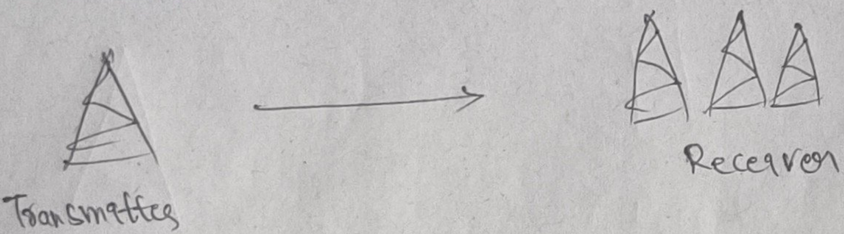
1. SISO – (Single Input Single Output)

→ 1 antenna on transmitter (1 antenna → 1 antenna)
→ 1 antenna on Receiver side → used in old system

2. SIMO – (Single Input Multiple output)

1 antenna → Multiple antennas

→ 1 antenna at transmitter
→ Multiple antennas at receiver



3. MISO – (Multiple Input Single output)

→ Multiple antennas at transmitter
→ 1 antenna at receiver

– multiple signals sent
– improves signal strength

Multiple antennas → 1 antenna

4. MIMO

Multiple antennas → Multiple antennas
Transmitter   Receiver

Working

→ Data is divided into multiple streams
→ Each stream is sent through different antennas
→ Receiver combines them to get original data
⇒ It $\uparrow$s speed, reliability

Techniques in MIMO

1. Spatial Multiplexing

→ Sends multiple data streams simultaneously.
→ $\uparrow$es data rate.

2. Beamforming – focuses on signal strength
3. Diversity Technique : improves signal reliability

Adv

→ High data rate
→ Better signal quality
→ more coverage
→ reduced errors

Dis

→ complex design
→ High cost
→ more Power
→ Signal processing is difficult.

Applications

- 5G networks
- Wi-fi
- LTE (4G)
- Satellite communication